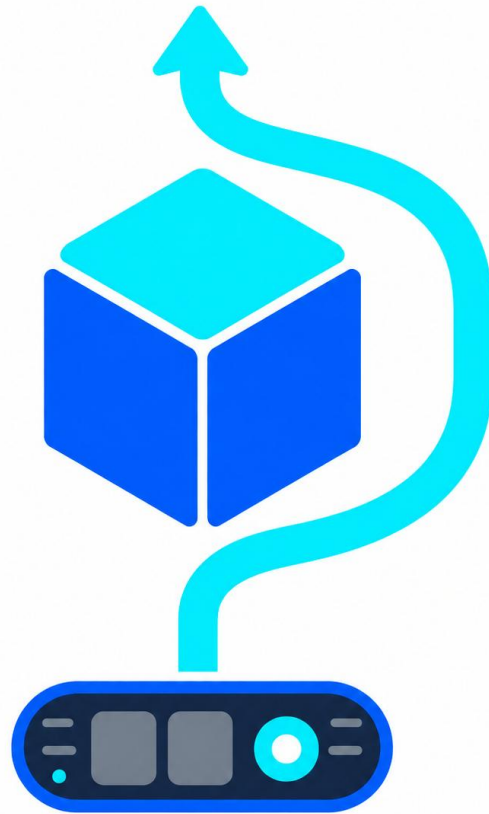




Obstacle Avoidance Solution

User Manual

3D sensing, zone detection and obstacle status output

Zhejiang MRDVS Technology Co., Ltd.

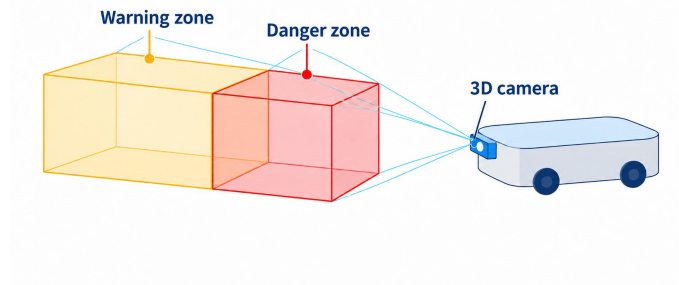
V0.1 · 2026-09-21

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1 Product overview

The Obstacle Avoidance Solution detects obstacles in zones you define. It sends the results to your control system, which can slow or stop the vehicle, or plan a route around an obstacle.



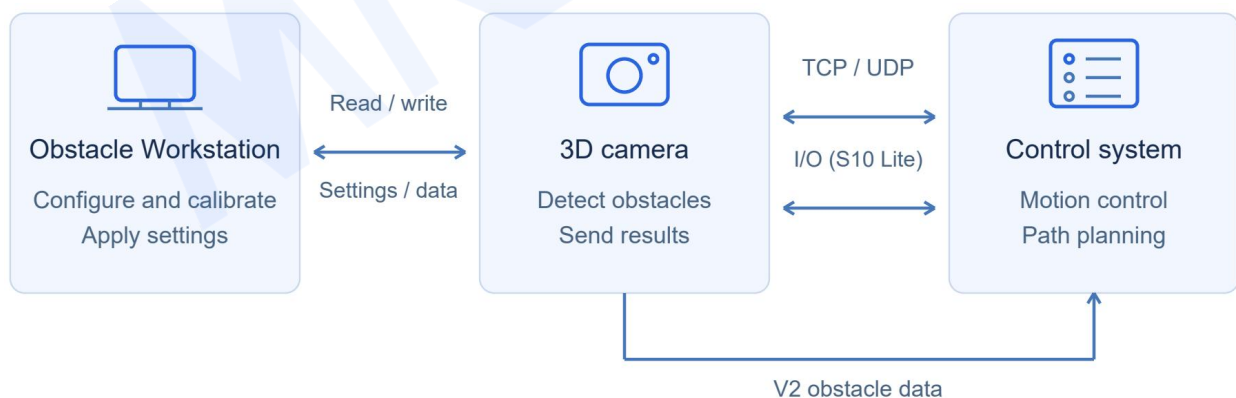
Danger and warning zones in 3D

Shared capabilities and supported cameras

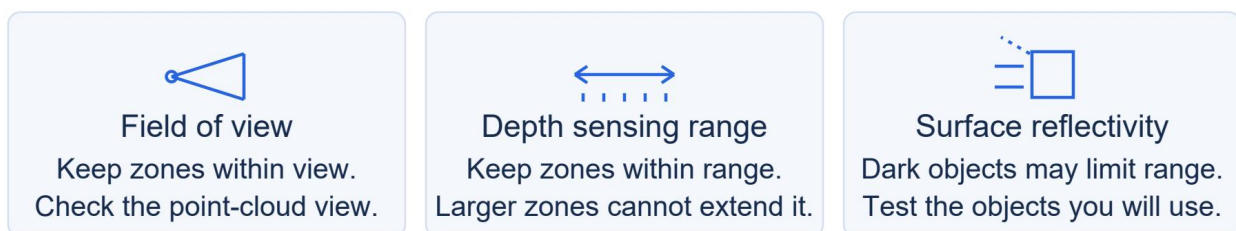
V1 and V2 share the same zone detection capabilities and algorithm performance. Both report Safe, Warning and Alarm. V2 adds obstacle bounding boxes and depth point-cloud output. S10 has RGB but no physical I/O; S10 Lite has physical I/O but no RGB.

V1 / V2	Supported cameras	Capabilities
V1	S10, S10 Lite, S11	Zone status output.
V2	S10 Pro	Zone status output, plus obstacle bounding boxes and depth point-cloud output.

How the system works

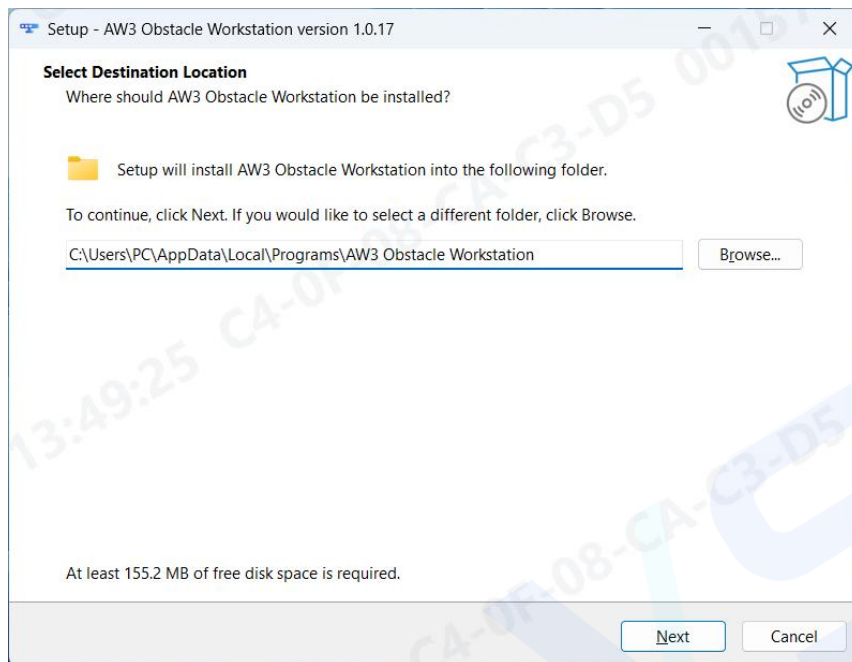


Camera limits



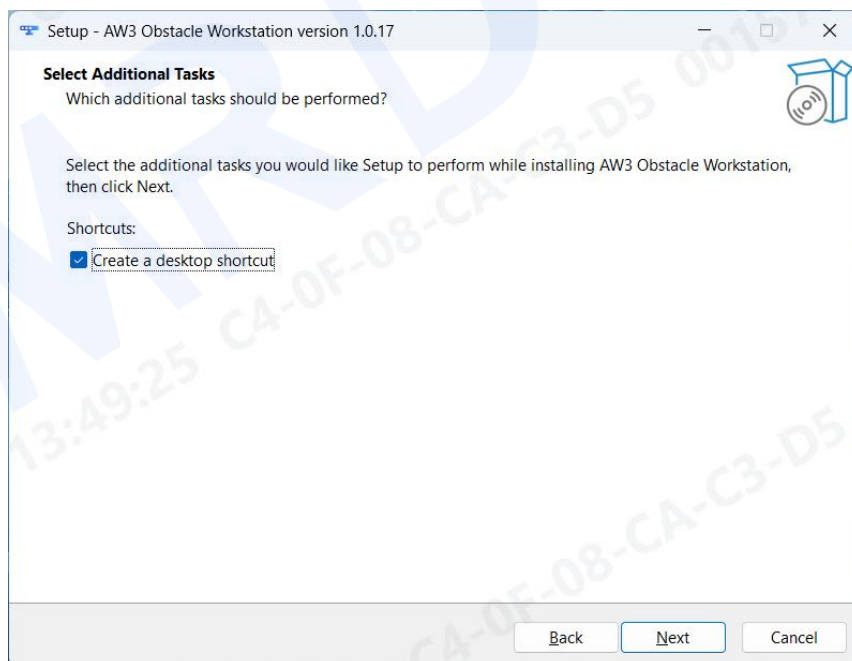
1.1 Install the workstation

1. Run the Obstacle Workstation installer.
2. Use the default installation folder, or click Browse to choose a different one. Click Next.



Select the destination folder

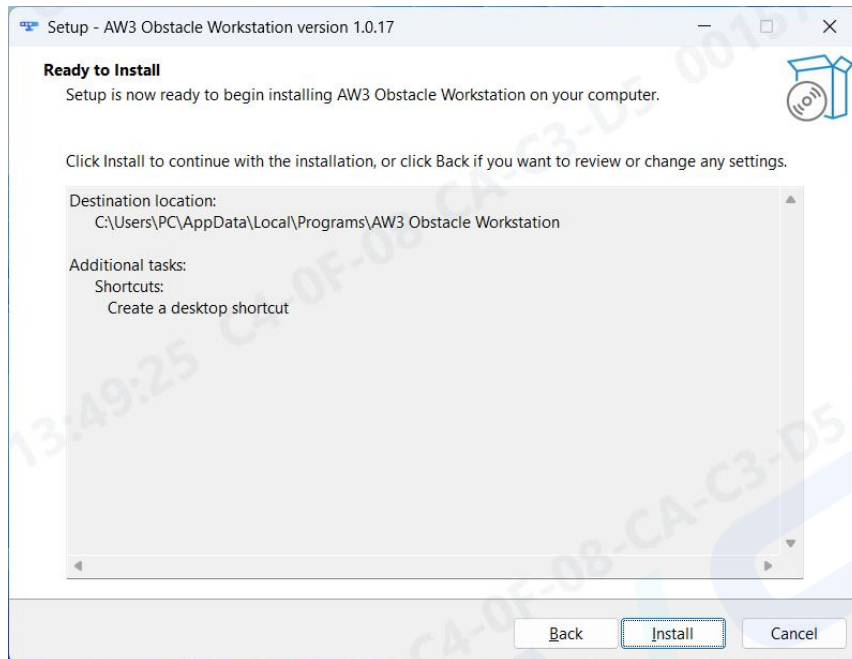
3. Select Create a desktop shortcut, then click Next.



Create a desktop shortcut

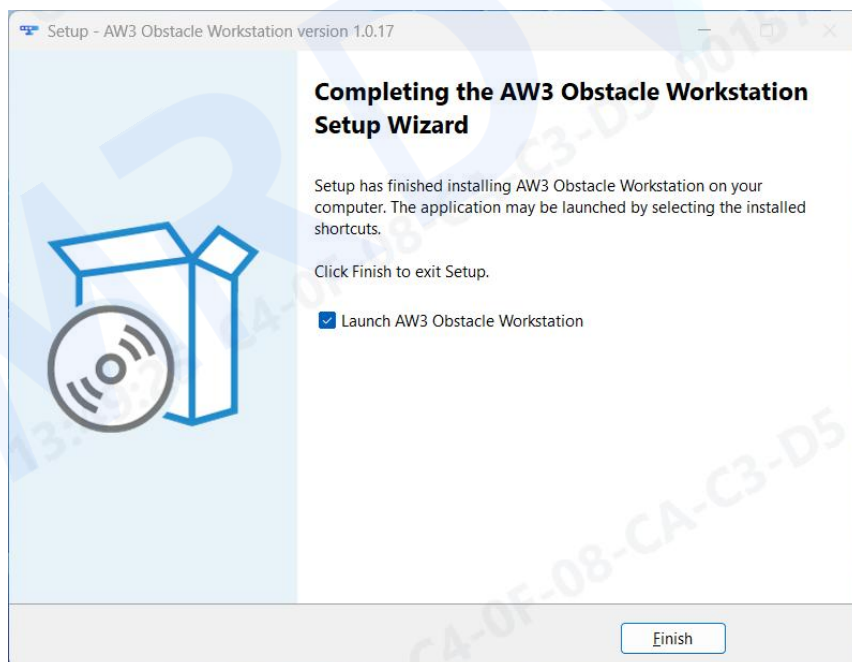
Complete installation

4. Check the settings and click Install.



Review the settings and install

5. Leave the launch option selected and click Finish to open the workstation.



Finish installation and launch the workstation

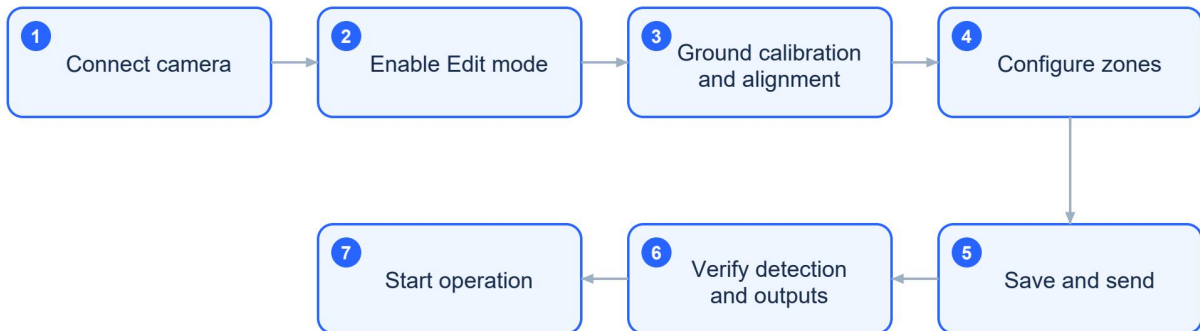


Launch from the desktop

Double-click the Obstacle Workstation shortcut to open the application, then connect the camera.

1.2 Setup at a glance

Follow these steps before putting the system into service.



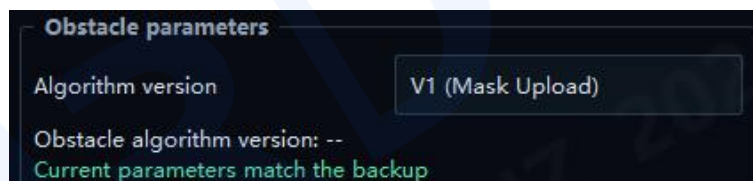
Initial setup workflow

Make your changes take effect

Click Save and send to apply changes to zones, camera alignment or output options. Do the same after importing settings, restoring defaults or calibrating the ground. I/O mode changes take effect automatically.

Check V1 / V2

The workstation identifies V1 or V2 from the connected camera. The result appears in the Algorithm version field under Obstacle parameters.



V1 / V2 in the configuration panel

Check that V1 / V2 matches your camera. This label identifies camera compatibility and output features, not a difference in zone detection performance.

2 Before you start

2.1 Mount the camera

Secure the camera where it has a clear view of the detection area. Keep the vehicle body, mounting brackets and housings out of view.

Low-mounted camera



Mount the camera at the front of the vehicle, facing the direction of travel. Keep its view clear of brackets and housings.

High-mounted camera on a forklift



Mount the camera on the mast, angled 30°–55° downward. A height of 2 m or less is recommended. Adjust the view to cover the area immediately in front of the forklift.

2.2 Check the installation

Check	Requirement
Power and network	Check the power supply and network cables. Make sure the camera and computer are on the same subnet.
View and live data	Make sure the camera can see the detection area and the live images are updating.
Ground	Use a clean, flat floor. Remove objects from the floor in the camera view before calibration.
Configuration	Apply your settings and confirm that the correct template is active.
Verification	Test safe, warning and alarm states. Check the results at the receiving controller.

What the Safe status tells you

Safe · 0 means no obstacle has triggered a warning or alarm within the configured zones. This status does not cover areas outside those zones, blocked from view or missing valid depth data.

2.3 Set the computer IP address

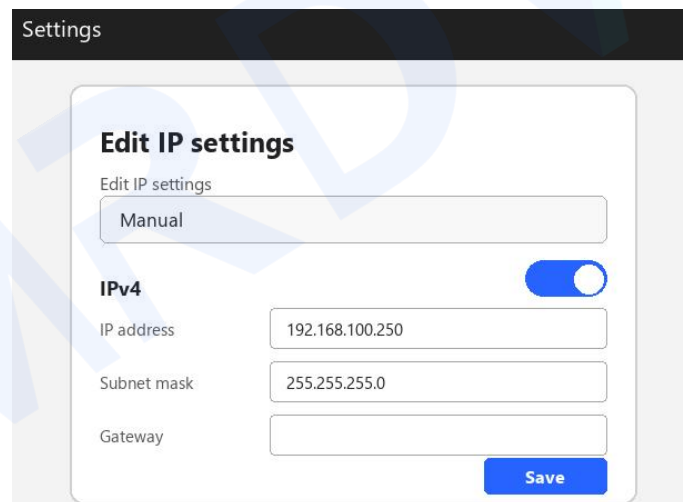
Give the camera and its computer network adapter different IP addresses on the same subnet. Change only the adapter connected to the camera, and record its existing settings first.

Device	Example address	Subnet mask
Camera	192.168.100.90	255.255.255.0
Computer adapter	192.168.100.250	255.255.255.0

With subnet mask 255.255.255.0, the first three numbers in the IP addresses must match; the last number must be different. The addresses shown are examples.

Windows

1. Open LxCameraViewer for Windows, search for cameras and check the camera IP address under Device Information.
2. Open Settings > Network & internet > Ethernet. Select the adapter connected to the camera, then click Edit next to IP assignment.
3. Select Manual, enable IPv4, and enter the IP address and Subnet mask. Use an unused address on the camera subnet. If Windows asks for a prefix length, enter 24 for 255.255.255.0.
4. Click Save and confirm that the new IP address is active.



Windows: edit the IPv4 settings of the camera network adapter

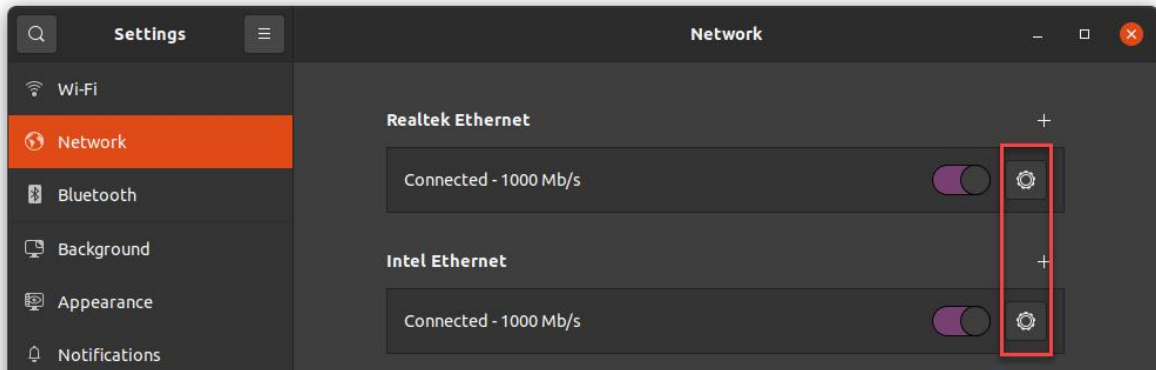
If the adapter is used only for the camera, leave Gateway and DNS blank. Otherwise, use the settings required by your network.

Return to the workstation, click Refresh devices and reconnect. Check that you have the right camera and that its live images and frame number are updating.

Linux computer IP settings

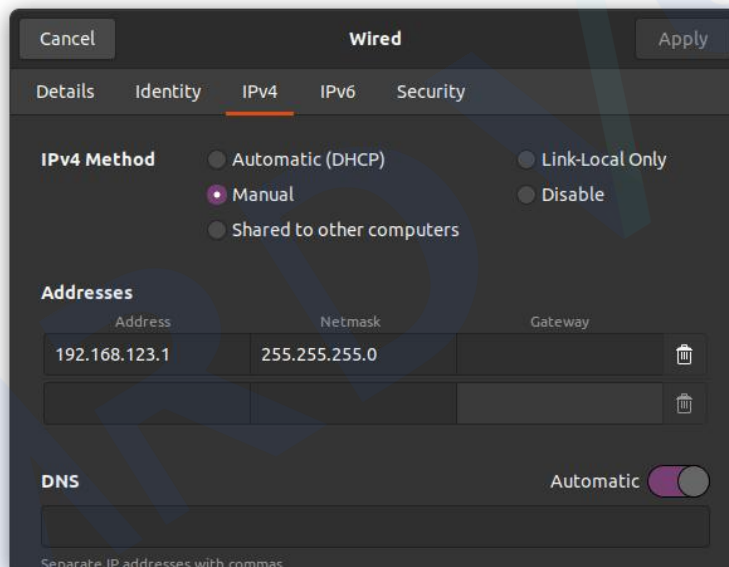
This example uses Ubuntu's wired network settings.

1. Open Settings > Network. Click the gear icon for the wired adapter connected to the camera.



Open the network adapter settings

2. Select IPv4 > Manual. Enter an unused Address and the Netmask for the camera subnet.



Manual IPv4 settings in Ubuntu

Use an available IP address on the same subnet as the camera. The address shown is an example.

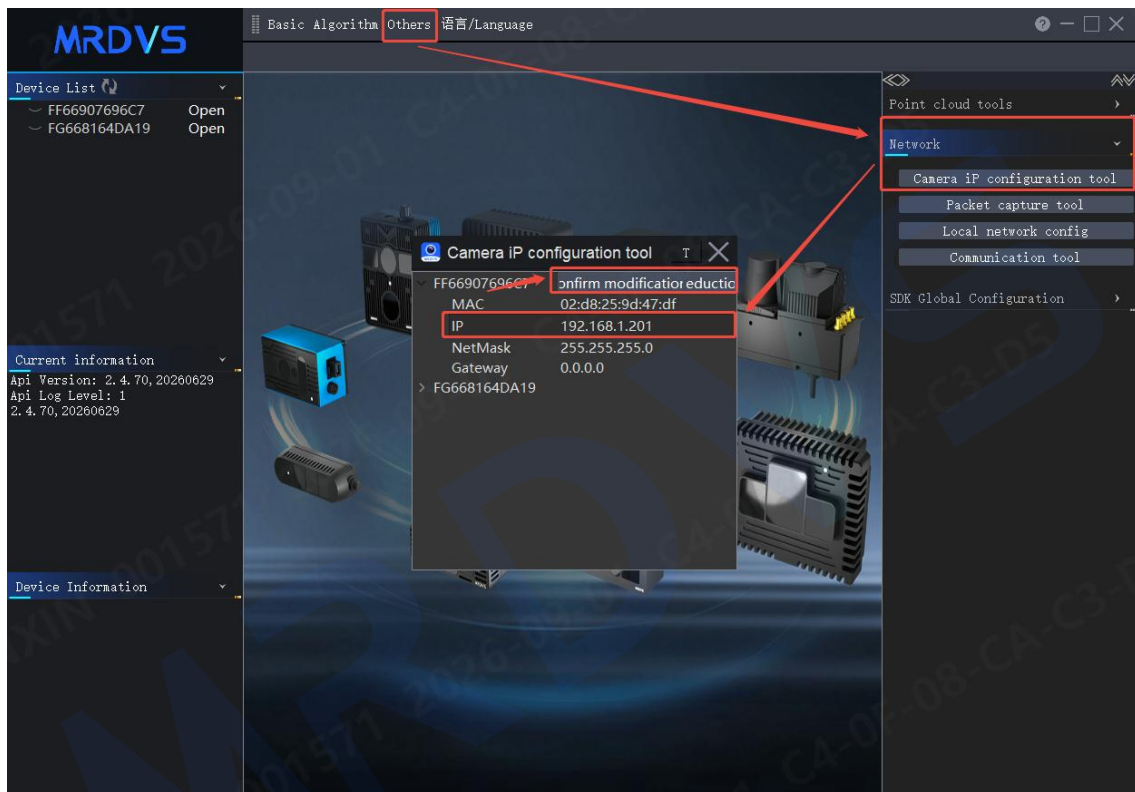
3. Click Apply, then reconnect the wired adapter. Confirm the new address before reconnecting the camera.

Leave gateway and DNS blank if the adapter is used only for the camera. Otherwise, use your network's settings.

2.4 Change the camera IP address

Use LxCameraViewer for Windows to change the camera's IP address, for example to resolve an address conflict or connect it to a different subnet.

1. Disconnect the target camera in the workstation. Open LxCameraViewer, search for cameras and identify the target by its camera ID.
2. Open Others > Network > Camera IP configuration tool.
3. Expand the target camera. Enter the new IP address and subnet mask; enter a gateway if required by the site network.



LxCameraViewer: change the camera IP address

4. Click Confirm modification for the selected camera and confirm that the new address has taken effect.
5. Close LxCameraViewer. Return to the workstation, click Refresh devices and reconnect. Confirm that the live images update.

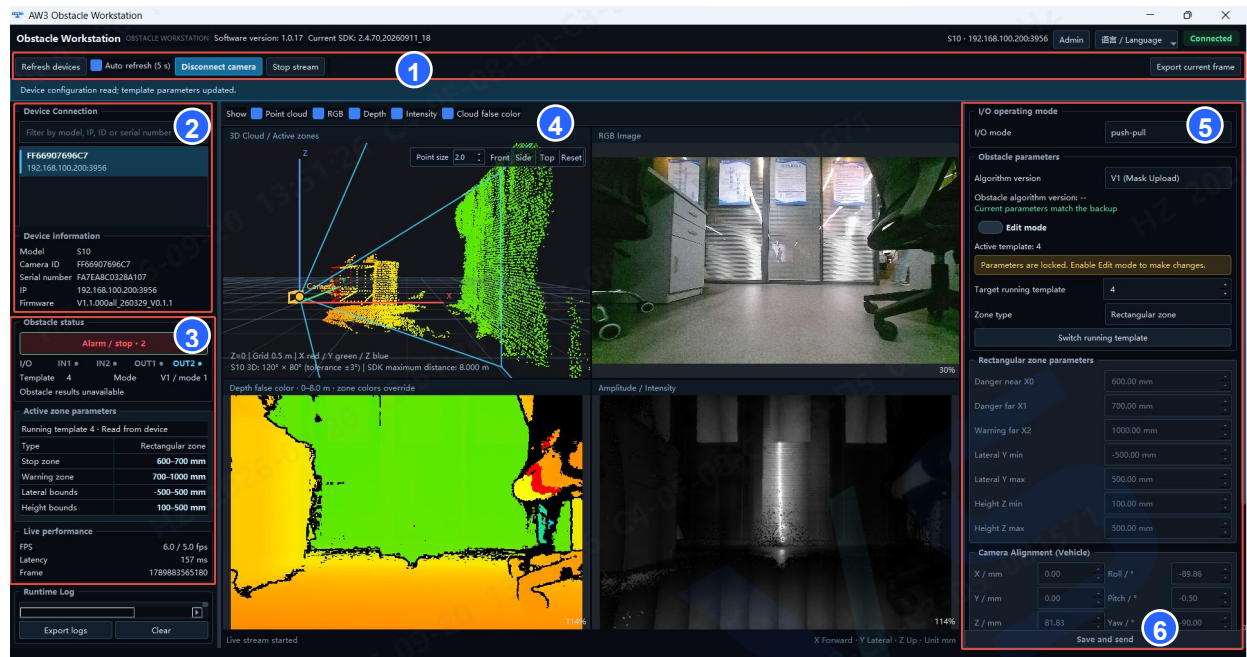
Keep the camera and computer adapter on the same subnet, with different IP addresses. If you move the camera to another subnet, update the adapter's address before reconnecting.

When setting up several cameras, change one address at a time. Use your network's address plan, not the example addresses shown.

3 Initial setup

3.1 Main interface

Use the numbered areas below to find the main controls.

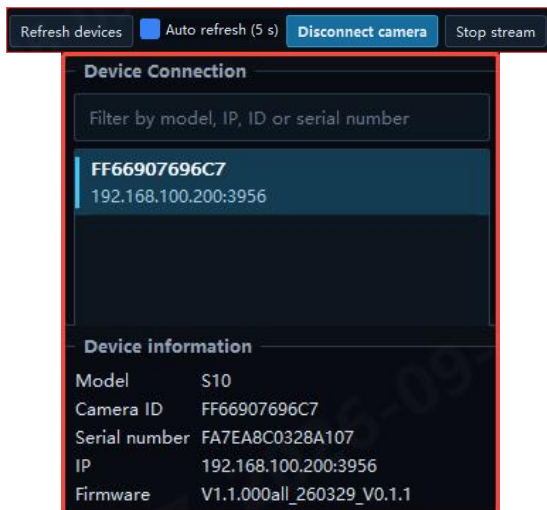


Main controls and display areas

No.	Area	Use
1	Toolbar	Find and connect to cameras, start or stop streaming, and export the current frame.
2	Device information	Select the camera and check its model, ID and connection information.
3	Status and runtime data	Check obstacle status, I/O signals, active zones and performance.
4	Image area	View point cloud, RGB, depth and intensity images, and preview changes to zones.
5	Configuration	Set zones, camera alignment, I/O and output options.
6	Save and send	Send your changes to the camera and make them take effect.

Runtime Log, below Live performance at the lower left, records workstation activity and errors.

3.2 Connect and check live data

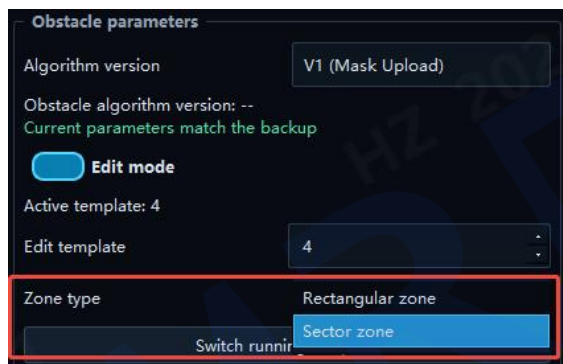


Device list and connection controls

1. Click Refresh devices and select the camera in Device Connection. Filter by model, IP, ID or serial number if needed.
2. Check that Connected appears at the upper right, the device details are correct and the images continue updating.

Auto refresh (5 s) refreshes the device list. If streaming is paused, resume it before checking detection results.

3.3 Edit mode



Edit mode and template selection

1. Enable Edit mode and select Edit template. Set the zone type and parameters while checking the point-cloud preview.
2. Click Save and send. Wait for confirmation before selecting another edit template or testing detection.

Mode	Displayed parameters and results
Off	The right panel shows the active template's settings and follows any template changes. The point cloud shows the active detection zone.
On	The zone preview changes as you edit, even before you apply your changes. Results on the left still use the active settings until Save and send succeeds.

Apply changes before editing another template

Click Save and send before choosing another template to edit, or your changes will be lost. The zone preview shows your draft settings, not detection results for those settings.

3.4 Align the camera and calibrate the ground

Camera alignment sets the camera's position and orientation in the installation coordinate system. All templates use the same alignment. X points forward, Y sideways and Z up. Position is in millimeters; angles are in degrees.

Camera Alignment and ground calibration controls

1. Fix the camera in its operating position. Use a clean, flat floor and remove objects from the visible ground.
2. Enable Edit mode. Manually adjust Z, Roll and Pitch until the ground point cloud is close to the Z=0 plane and approximately level. This initial alignment improves calibration accuracy.
3. Click ground calibration. The calibration calculates Z, Roll and Pitch. Check that the ground point cloud is level and near Z=0; if not, check the floor and repeat the initial alignment and calibration.
4. Set X, Y and Yaw to match the actual mounting position and detection direction. Use Top view to check the horizontal alignment.
5. Click Save and send. Wait for confirmation, then verify the detection states.

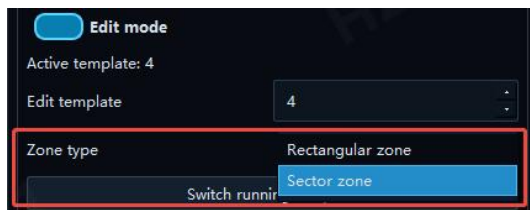
Parameter	Meaning
Z / Roll / Pitch	Camera height, roll and pitch. Adjust them manually before running ground calibration.
X / Y	Forward and sideways offsets from the installation origin.
Yaw	The camera's horizontal angle relative to the detection direction.

Apply and verify the calibration

Calibration updates the values on screen. Click Save and send to apply them to the camera. Repeat alignment after moving or replacing the camera, or using Reset extrinsics.

3.5 Configure templates and zones

You can save 20 sets of zone settings as templates 0–19. Each template has its own zone type and dimensions; camera alignment is shared across all templates.



Turn on Edit mode, choose a template under Edit template, then select the Zone type.

Zone positions are relative to the installation origin: X forward, Y sideways and Z up. Distances are in millimeters.

3.5.1 Rectangular zone

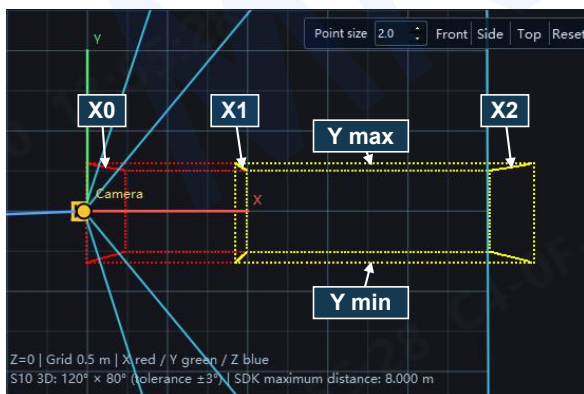
The danger zone extends from X0 to X1, and the warning zone from X1 to X2. Points with an X coordinate below X0 are outside the configured zone.



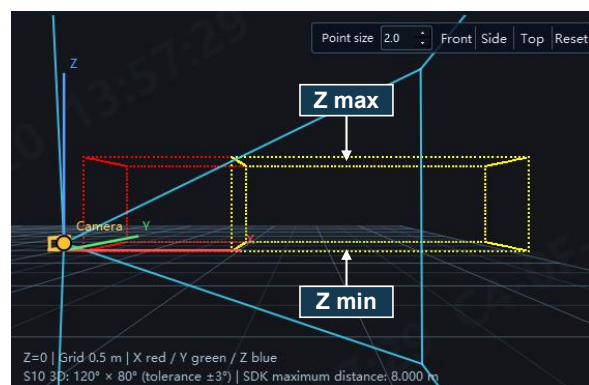
Rectangular zone parameters

Parameter	Meaning
Danger near X0	Start of the danger zone.
Danger far X1	Where the danger zone ends and the warning zone begins.
Warning far X2	End of the warning zone.
Lateral Y min / Y max	Left and right limits.
Height Z min / Z max	Minimum and maximum detection height.

Required: $X0 < X1 < X2$; $Y \min < Y \max$; $Z \min < Z \max$.



Top view: distance and width



Side view: height limits

The red outline marks the danger zone; the yellow outline marks the warning zone. Points outside the width or height limits are ignored.

X values are measured from the installation origin. To find the distance from the camera to an X boundary, subtract the camera's X position from that boundary's X value.

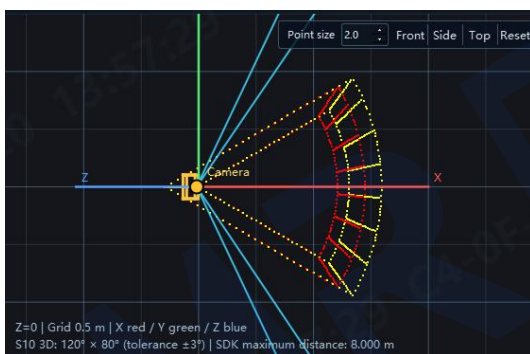
3.5.2 Sector zone

The danger zone extends from the center to Danger radius. The warning zone extends from there to Warning radius. Set the angles and height limits to define the rest of the zone.

Sector zone parameters

Parameter	Meaning
Danger radius	Distance from the center to the edge of the danger zone.
Warning radius	Distance from the center to the outer edge of the warning zone.
Left / Right boundary angle	Angles that define the left and right edges.
Height Z min / Z max	Minimum and maximum detection height.
Center X / Center Y	The center's X and Y coordinates.

Required: danger radius < warning radius; left angle < right angle (-180° to 180°); Z min < Z max.



Check the sector in the point cloud

Checks for either zone type

Coverage: check the zone in the point cloud. Keep it within the camera's field of view and effective range.

Objects: test representative objects at the distances you expect. Check for missing depth data inside the zone.

Apply: click Save and send when the preview is correct. Wait for confirmation before editing another template.

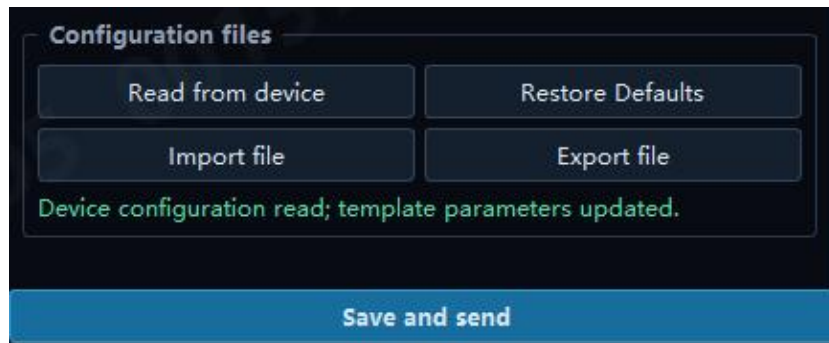
Setting Z min to 0 mm can cause false alarms

For either zone type, dust near the floor can trigger false alarms. Do not use 0 mm as a default for every installation. After applying your settings:

1. Clear the zones and check for false alarms under normal floor conditions.
 2. Place a low obstacle in the zone and check detection and the controller's response.
- If you raise Z min to reduce interference, repeat both tests to make sure low obstacles are still detected.

3.6 Save and apply the configuration

1. Check the edit template, zone, camera alignment and output options. Click Save and send and wait for confirmation.



Configuration controls and Save and send

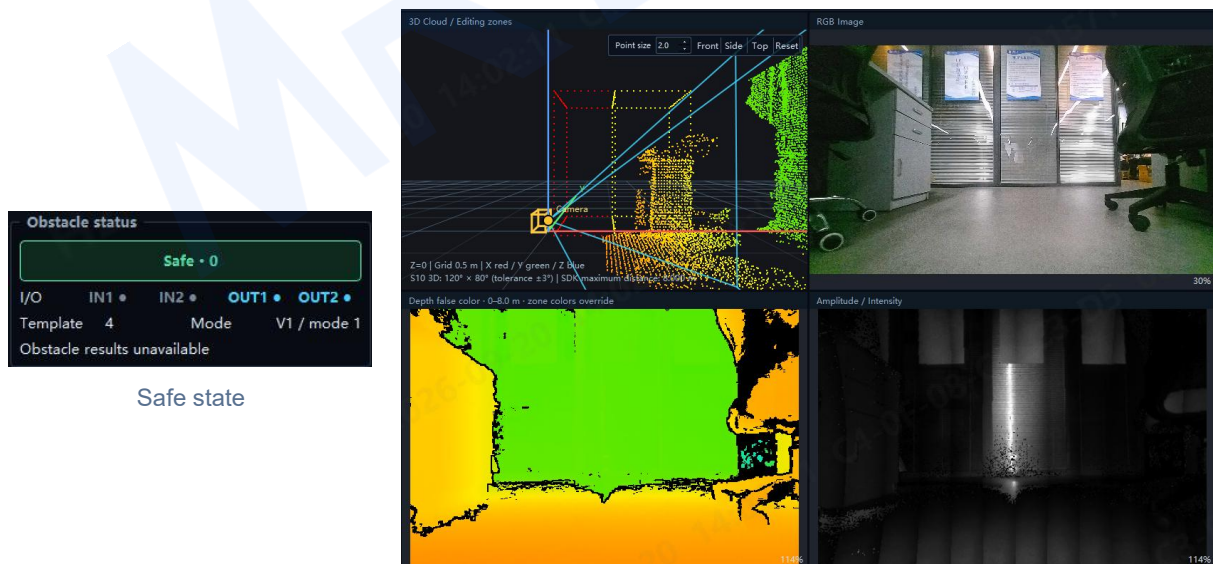
2. Confirm the active template and read the applied values in Active zone parameters on the left. If needed, use Switch running template to select the configured template.
3. Turn Edit mode off. Verify safe, warning and alarm states with live data. If sending fails, resolve the reported problem and send again.

3.7 Verify obstacle detection

Before each test, check the active template and zone. Use the four image views to inspect the scene. After changing Z min, test both an empty zone and a low obstacle.

3.7.1 Empty zones should report Safe

Remove obstacles from both zones. Expected result: Safe · 0. On S10 Lite, OUT1 / OUT2 = 1 / 1.



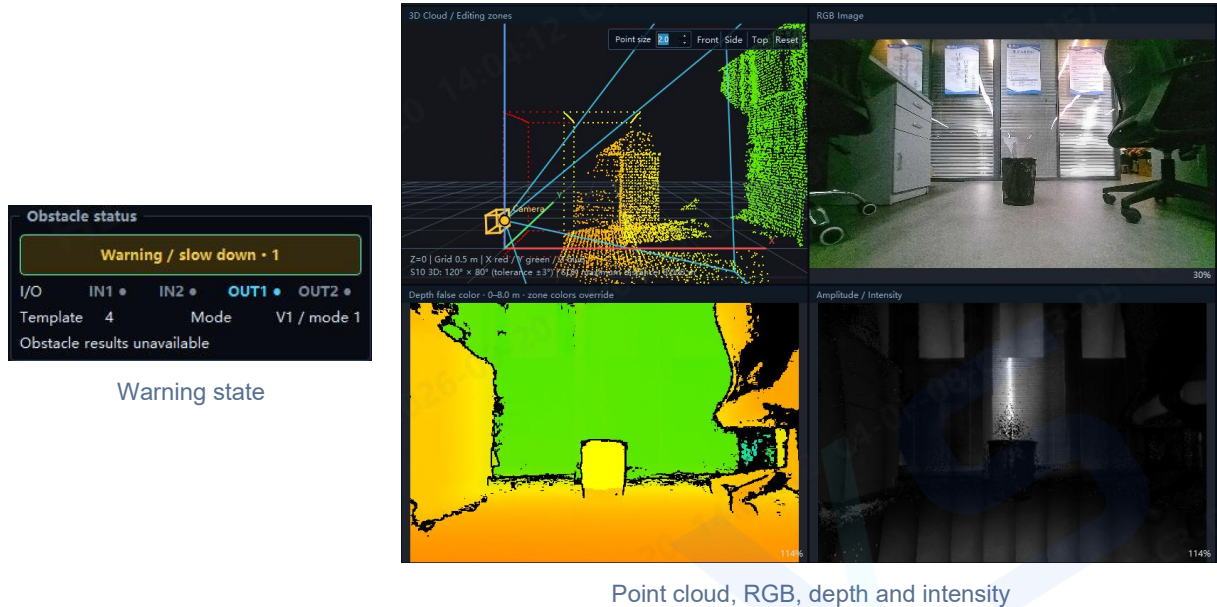
Safe state

Point cloud, RGB, depth and intensity

Verify warning and alarm

3.7.2 Obstacle in the warning zone

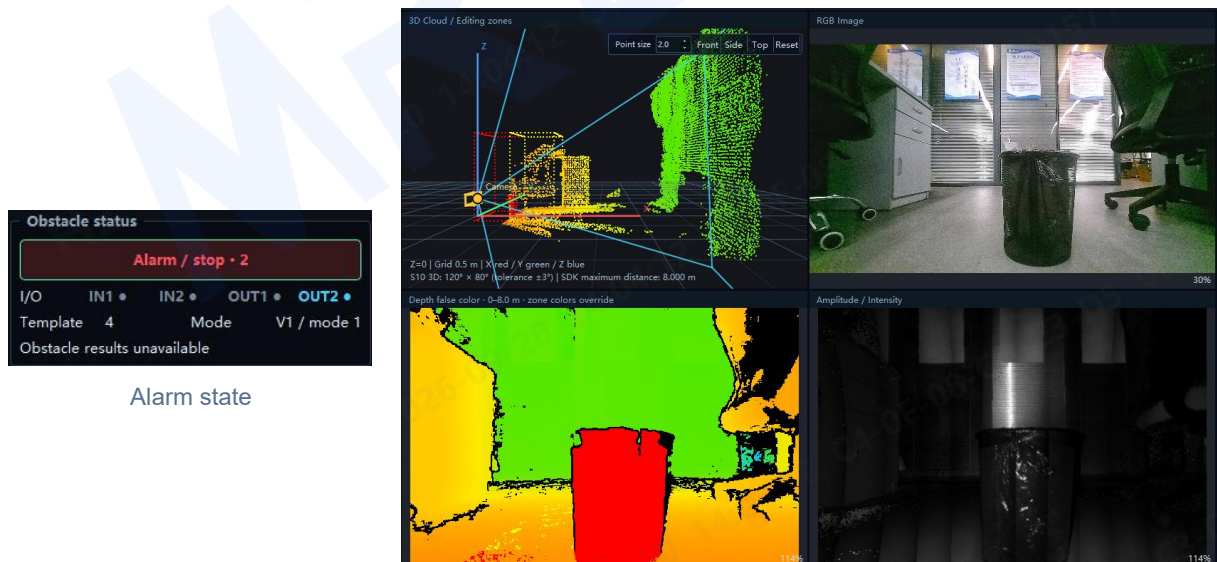
Place an obstacle in the warning zone. Expected: Warning / slow down · 1. On S10 Lite, OUT1 / OUT2 = 1 / 0.



Point cloud, RGB, depth and intensity

3.7.3 Obstacle in the danger zone

Move the obstacle into the danger zone. Expected: Alarm / stop · 2. On S10 Lite, OUT1 / OUT2 = 0 / 1.



Point cloud, RGB, depth and intensity

Remove the obstacle and check that the status returns to Safe in both the workstation and controller. For S10 Lite I/O, 1 means HIGH and 0 means LOW. In open-drain mode, the receiving circuit establishes HIGH.

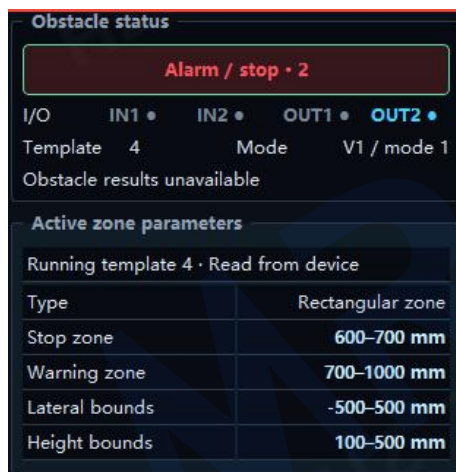
3.8 Daily operation

Startup checks

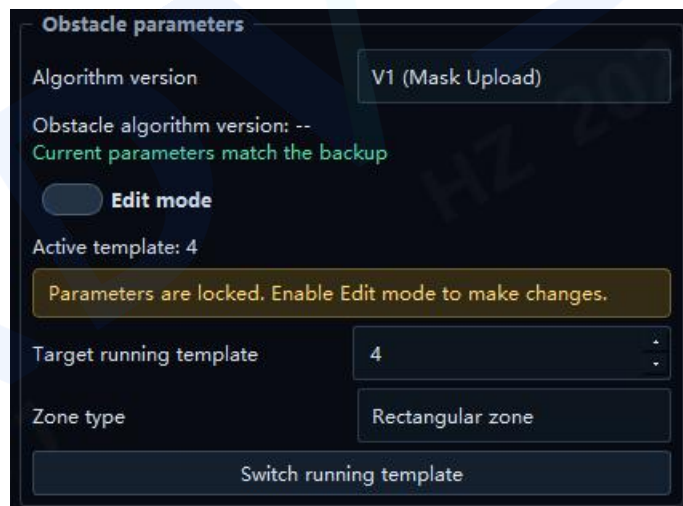
Check	Action
Connection and live data	Connect to the correct camera. Check that its live images and frame number are updating.
Running template	Turn off Edit mode. Check Active template and the settings shown under Active zone parameters on the left.
Obstacle status	Check the status in the workstation and controller. Retest safe, warning and alarm states after changing the installation or settings.

Switch the running template

1. With Edit mode off, set Target running template.
2. Click Switch running template. Wait until Active template shows the requested number.
3. Check the left-side zone parameters and current result. With Edit mode off, the right panel updates to show the new running template.



Status and active zone parameters



Running template controls

Check which template is active

Choosing a template number does not activate it. Click Switch running template, then check Active template to confirm the change.

4 Functions and parameters

4.1 View images and point clouds

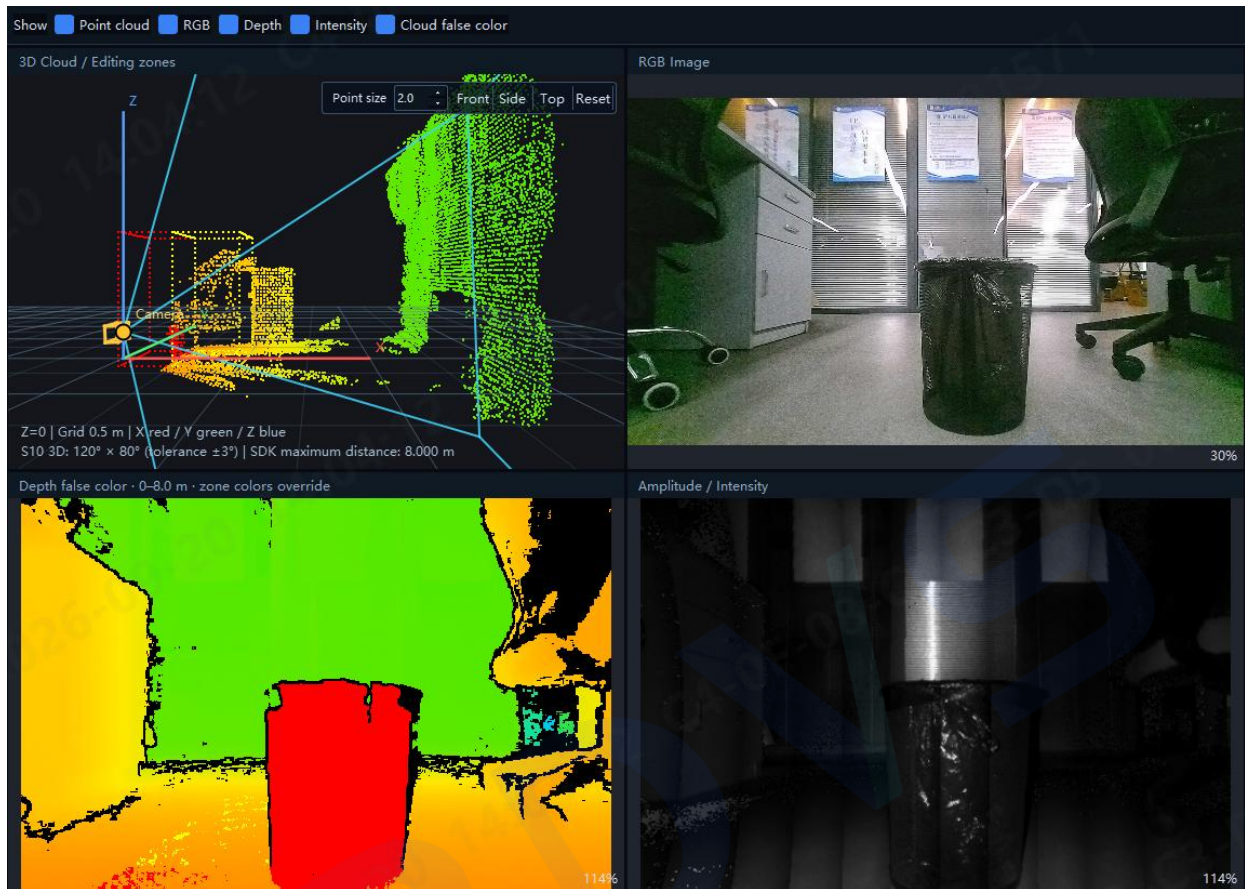


Image views and display controls

Display	Use
Point cloud	Check where the zones sit relative to the floor and obstacles.
RGB	Identify objects and see what may be blocking the view. S10 Lite has no RGB image.
Depth	Check depth data and highlighted detection zones.
Intensity	Inspect the amplitude or intensity image.
Cloud false color	Use color to distinguish detection zones and depth in the point cloud.

Use Top to check zone length and width, Side to check height, and Front for a front view. Click Reset to restore the view. Adjust Point size to make points easier to see.

Display controls do not change detection settings. To confirm detection is running, check that the frame number and live images are updating.

4.2 Manage configuration files

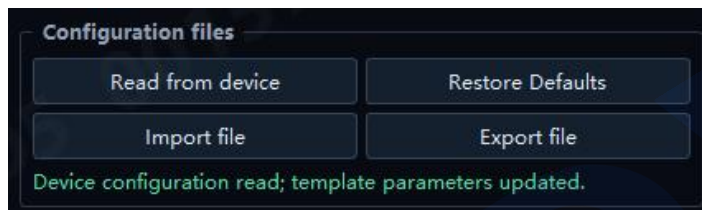
Scroll down the right panel to Configuration files.

Control	Action
Read from device	Load the camera's current settings into the workstation. Apply any edits you want to keep before using this control.
Export file	Save the settings as a JSON file.
Import file	Open a saved JSON file, check the settings, then click Save and send to apply them.

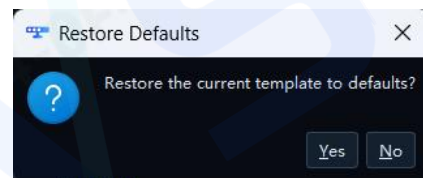
4.3 Restore template defaults

Restore Defaults resets the zone settings for the template you are editing. Camera alignment stays unchanged.

1. Enable Edit mode and select the intended template. Click Restore Defaults.



Configuration file controls



Restore Defaults confirmation

2. Adjust the default values for your installation, then click Save and send. Wait for confirmation and verify all three detection states.

Recalibrate after installation changes

After moving or replacing the camera, align it and calibrate the ground again. Importing saved settings does not replace these steps.

4.4 Obstacle bounding boxes and Simple output

These settings control V2 data output. Simple output returns only status and the bounding-box count.

Option	Effect
Obstacle bounding boxes	Show the number of obstacles and mark detected obstacles with bounding boxes.
Simple output	Output only the obstacle status (I/O result) and bounding-box count.

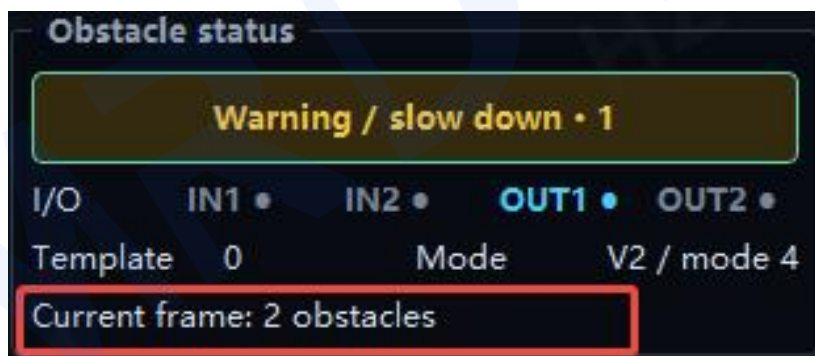


Output options: Obstacle bounding boxes and Simple output

1. Enable Edit mode and choose the required output options.
2. Click Save and send. Wait for confirmation, then check the count and displayed boxes.

Get bounding-box details

Select Obstacle bounding boxes, clear Simple output, then click Save and send.



Example: two obstacles detected

5 Status and communication

5.1 Read obstacle status

Status	Meaning	S10 Lite OUT1 / OUT2
Safe · 0	No obstacle has triggered a warning or alarm in the configured zones.	1 / 1
Warning / slow down · 1	An obstacle is in the warning zone.	1 / 0
Alarm / stop · 2	An obstacle is in the danger zone.	0 / 1
Invalid status · 1000	A valid obstacle status is not available.	0 / 0
No valid output / --	A valid result has not been read.	Not read

1 = HIGH logic level; 0 = LOW logic level. The physical level depends on the I/O mode. Do not interpret an invalid status as Safe.



Invalid status 1000 and active zone information

If Invalid status 1000 appears, check the live stream, active template and whether your settings were applied. Export the workstation log. If No valid output appears, check the connection and stream.

Obstacle results unavailable means the workstation has not received obstacle details. It does not mean there are no obstacles. Check the main status indicator for Safe, Warning or Alarm.

5.2 Set the S10 Lite I/O mode

Physical I/O is available only on S10 Lite; S10 has no I/O. Screenshots retain older model labels and illustrate the controls only. Under I/O operating mode, choose open-drain or push-pull to match the controller's input circuit. The setting takes effect automatically.



Select open-drain or push-pull

Logic value	Open-drain	Push-pull
1 — HIGH	The output floats; the receiving circuit or a pull-up sets the HIGH level.	24 V
0 — LOW	0 V	0 V

The mode applies to both OUT1 and OUT2. After changing it, test safe, warning and alarm states and check the voltage levels at the controller inputs.

5.3 V2 output data

With Simple output selected, the camera returns only obstacle status (the I/O result) and the bounding-box count. For box details, clear Simple output and select Obstacle bounding boxes.

Output	Description
Obstacle result I/O	The obstacle status: safe, warning or alarm.
Bounding-box count	Number of obstacle bounding boxes in the current frame.
Bounding-box details	Box center, orientation, width, height and depth, plus the obstacle centroid.
Algorithm return status	0 means successful execution; a negative value means an error.
Depth point cloud (filtered)	Point count and data after ground and noise filtering.
Result time	Timestamp of the result.

Obstacle status and processing status are different

Obstacle status tells you whether the scene is safe or contains an obstacle. Algorithm return status tells you whether processing succeeded. Only S10 Lite has physical I/O outputs.

5.4 TCP and UDP communication

First configure your templates in the workstation and click Save and send. Your controller can then use TCP or UDP to receive obstacle status and switch templates. Keep the camera and controller on the same subnet. These messages do not include bounding boxes or point clouds.

Mode	Connection
TCP	Connect to the camera's IP address on port 6688 as a TCP client. The camera starts sending results once connected.
UDP	Bind a UDP socket to the controller's local IP address and listening port. Send the initialization message to the camera on port 6688, then receive results on that same socket.

Message reference

Direction and purpose	Hexadecimal bytes
Controller → camera: initialize UDP	AC ED 04 00 00 00 00 00
Camera → controller: safe	AC ED 04 10 XX 00 00 00
Camera → controller: warning	AC ED 04 10 XX 00 00 01
Camera → controller: alarm	AC ED 04 10 XX 00 00 02
Camera → controller: template unavailable ¹	AC ED 04 10 XX 17 03 E8
Controller → camera: switch template	AC ED 04 1A 00 00 00 NN
Controller → camera: save default template	AC ED 04 1B 00 00 00 00
Camera → controller: default saved	AC ED 04 1B 00 00 00 01

Each message is 8 bytes long. In a result message, byte 5 (XX) is the active template number. In a switch command, NN is the template to activate. Templates 0–19 are 00–13 in hexadecimal. TCP reads may contain part of a message or several messages; buffer incoming bytes and parse them in groups of 8.

¹ The template has not been applied to the camera, or its number is out of range. Apply a valid template and try again.

On S10 Lite, I/O inputs take priority over network commands when selecting a template. If a switch has no effect, check IN1 and IN2. Results are sent at the image frame rate; wait for confirmation before switching again.

Communication examples

This example activates template 4 and saves it as the startup default. Apply template 4 in the workstation first. Replace the example IP address with your camera's address.

1 Establish the connection

TCP: connect and receive	UDP: initialize and receive
Select TCP Client. Connect to 192.168.100.90:6688. Results arrive automatically.	Bind a local receive port. Send the bytes below to 192.168.100.90:6688. AC ED 04 00 00 00 00 00 Receive on the same local socket.

Set the communication utility to HEX for both send and receive. Transmit the bytes shown, not their text representation.

2 Select template 4 and verify the result

Send: select template 4	Receive example: safe
AC ED 04 1A 00 00 00 04	AC ED 04 10 04 00 00 00

Wait for a valid result with 04 in the fifth byte. This confirms that template 4 is running. The final byte in a valid result is 00 for safe, 01 for warning or 02 for alarm.

3 Save the selection for the next startup

Send: save current selection	Receive: saved successfully
AC ED 04 1B 00 00 00 00	AC ED 04 1B 00 00 00 01

Check for the confirmation message. If you skip this step, the camera returns to its previous default template after a restart.

If the expected result does not arrive

Check the connection first; for UDP, also check that you sent the initialization message. If the template does not change, check the I/O inputs and make sure the template was applied. An invalid result must not be treated as Safe.

6 Troubleshooting and maintenance

6.1 Connection and data issues

Before changing detection settings, check power, connections and the live stream. After fixing the issue, check the results in both the workstation and controller.

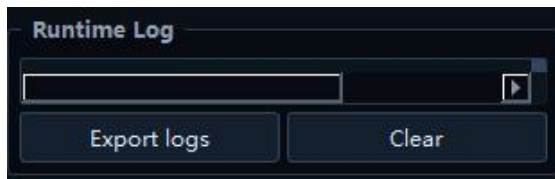
Symptom	Action
No devices listed	Check power and cables. Confirm that the camera and computer are on the same subnet. Clear filters and click Refresh devices.
Wrong camera connected	Check ID, serial number and IP. Disconnect and select the intended camera.
Images stop updating	Resume the stream and check the frame number. Reconnect if necessary.
No valid output	Restore the connection and live stream, then recheck the result.
Invalid status 1000	Check the live stream, active template and whether the settings were applied. Export the workstation log. Do not treat an invalid result as Safe.
TCP or UDP receives no results	Check the camera's IP address and port 6688. For TCP, check the connection. For UDP, send the initialization message and keep the receiving socket open.

6.2 Configuration and detection issues

Symptom	Action
Fields are disabled	Enable Edit mode and select Edit template.
Changes or a template switch do not take effect	Confirm that Save and send succeeded and check Active template. On S10 Lite, check IN1 and IN2: a template selected by I/O takes priority.
Edits disappear after switching templates	Re-enter your changes and click Save and send before choosing another template to edit.
Save or send fails	Check the connection and make sure the parameter values meet the required limits. Correct any errors, then try again.
Alarm when no obstacle is expected	Check whether the zone includes the floor, vehicle body or a bracket. If the ground point cloud appears tilted, align it manually, recalibrate on a clean, flat floor, and click Save and send.
Black object has few or no depth points	Check the depth image and point cloud. Dark surfaces reflect less light. Where practical, reduce the distance or add a higher-reflectivity marker to the object, then test again.
Low obstacle is not detected	Check Height Z min, ground calibration and camera mounting height. A setting of 0 mm may pick up ground dust. Apply your changes, then test an empty zone and a low obstacle.
Points are in the zone but the state is unexpected	Check the active template, zone limits and camera height. Read the settings from the camera, then test with an obstacle at a known position.
Mounting structure distorts the data	Check for brackets or housings in the view, a recessed camera, protective film left on the lens or added cover glass. Correct the installation and repeat calibration.
Status changes but controller input does not	For S10 Lite I/O, check output mode, grounding and wiring. For TCP/UDP, check the received template number and status bytes.

The zone preview updates as you edit. Results on the left continue to use the active settings until your changes are successfully applied.

6.3 Export logs and the current frame



Workstation log



Export current frame

Function	What it exports
Export logs	Records workstation activity and software errors, not events inside the camera. Choose where to save the log.
Export current frame	Export point cloud, RGB, depth and intensity images. RGB images require an RGB-equipped model. Confirm that the live stream is updating before exporting.

Both features are available in V1 and V2. Export any logs you need before clicking Clear, which removes the entries shown in the workstation.

6.4 When to repeat setup and testing

Change	Required action
Camera or mounting position changes	Repeat camera alignment and ground calibration. Save and send, then verify all states.
Zone type or bounds change	Save and send. Select the template and verify its active bounds and results.
Output options change	Save and send, then inspect the output and box count/details.
Configuration imported or defaults restored	Save and send, then verify safe, warning and alarm.
S10 Lite I/O mode changes	The mode is sent automatically. Verify levels at the receiving input.
Software or firmware changes	Reconnect, check live data and verify status and outputs.
Floor or obstacle type changes	Recalibrate if the floor changes. Test with the changed obstacle type.

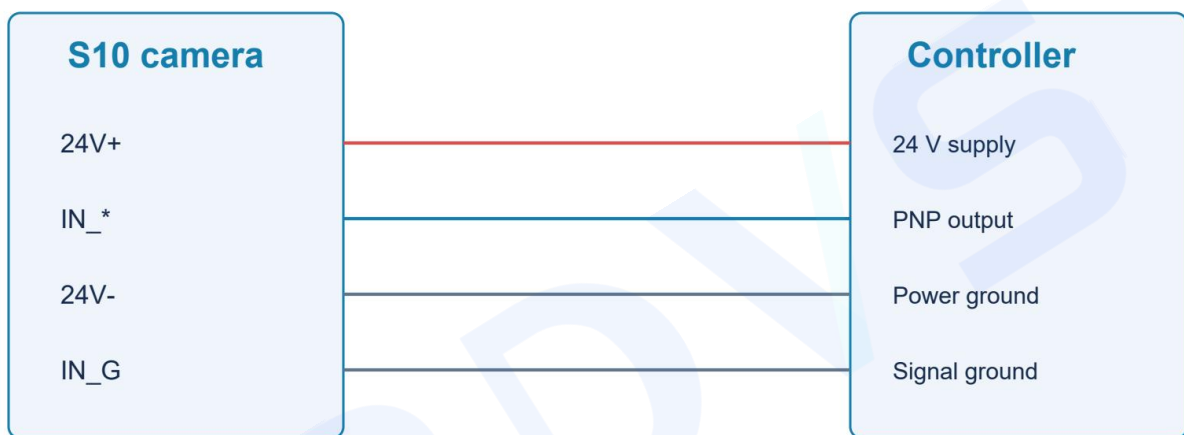
Appendix A S10 Lite I/O wiring

These wiring instructions apply only to S10 Lite. The older S10 labels in the diagrams refer to S10 Lite. The diagrams identify connections by terminal name. Match them to the labels on your S10 Lite cable; pin numbers and wire colors from other models may differ.

A1 Power and common ground

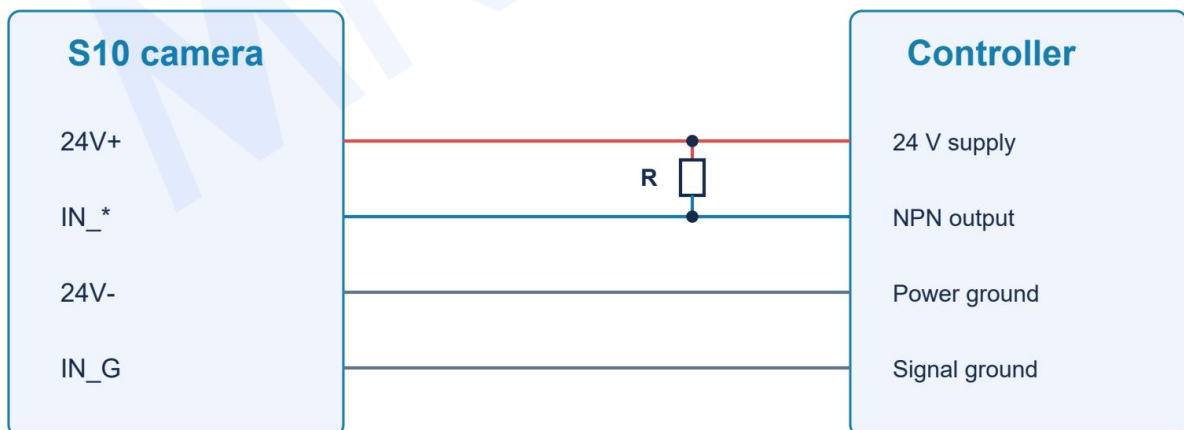
Disconnect power before wiring. Connect the camera's 24V- to the controller's power ground, and IN_G and OUT_G to its signal ground. If the controller uses one common ground, connect all three to GND.

A2 Controller output to camera input



* = 1 or 2. Terminal labels are shown; cable colors and pin numbers are not specified.

PNP controller output connected to S10 Lite IN1 or IN2



* = 1 or 2. Terminal labels are shown; cable colors and pin numbers are not specified.

NPN controller output with a pull-up when required

In the NPN circuit, resistor R pulls the signal up to 24 V. Whether you need it depends on the controller's output circuit. Check the controller specifications and test the input wiring.

In these diagrams, red means 24 V, blue means signal and gray means ground. Wire by terminal label, not by color alone.

A3 S10 Lite outputs and template inputs



* = 1 or 2. Terminal labels are shown; cable colors and pin numbers are not specified.

S10 Lite OUT1 or OUT2 connected to a controller digital input

Choose an output mode that matches the controller input, then test the status-to-output mapping. Both outputs LOW may mean obstacle detection is disabled, starting up or in a fault state. It does not mean Safe.

Select a template through inputs

IN1	IN2	Template selection
1 — HIGH	1 — HIGH	Template 2
1 — HIGH	0 — LOW	Template 1
0 — LOW	1 — HIGH	Template 0
0 — LOW	0 — LOW	Default or temporarily selected template

An unconnected input reads LOW. The recommended voltage for HIGH is 5–25 V. After changing the inputs, check Active template in the workstation.

Which template takes priority?

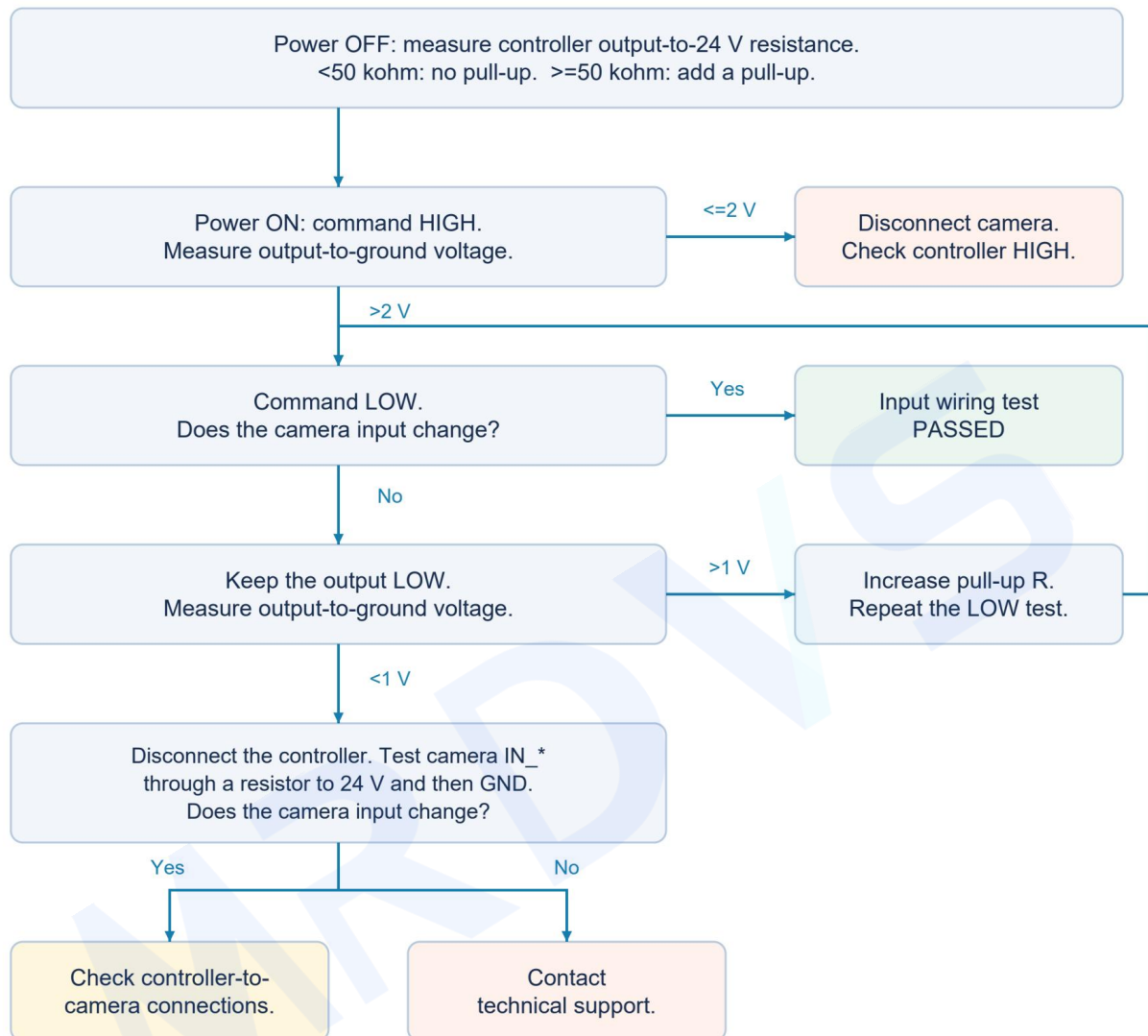
When IN1 and IN2 select template 0, 1 or 2, that selection takes priority. Workstation and network commands can change the stored or temporary selection, but cannot override the template selected by the inputs.

When both inputs are LOW, the latest stored or temporary selection is used. After a restart, the camera uses the saved default. A TCP or UDP selection lasts only until restart unless you save it as the default.

Wait for Active template to show the requested number before testing the new zone. Do not keep sending switch commands while you wait.

A4 Check S10 Lite input wiring

Use these checks to confirm that the S10 Lite input responds to the controller output. Disconnect power before measuring resistance.



S10 Lite input wiring checks

If a pull-up is required, use the recommended 10 kΩ through-hole metal-film resistor rated for at least 1 W. Do not use less than 3 kΩ and do not connect the input signal directly to 24 V.

The 2 V and 1 V thresholds are for troubleshooting, not the complete input specification. Contact technical support if a reading is unstable or close to a threshold.

Before putting the system back into service, test template selection through the inputs and check the controller's response to safe, warning and alarm states.

Revision history and technical support

Version	Date	Revision
V0.1	2026-09-21	Updated interface examples, V1/V2 functions, camera operating limits, calibration and S10 Lite wiring.

Technical support

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Website: mrdvs.com

GitHub: <https://github.com/Lanxin-MRDVS>

Knowledge Community: <https://docs.mrdvs.com/en>

LinkedIn: <https://www.linkedin.com/company/mrdvs>



Scan to open MRDVS on GitHub

When contacting support, include the camera ID, software and firmware versions, active template, a description of the problem, the workstation log and an exported frame.